

MILAN TIWARI

M.S. Robotics and Autonomous Systems — Embodied AI, Computer Vision, 3D Scene Understanding — Available Summer 2026
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EDUCATION

- **Arizona State University**, Tempe, AZ Aug 2024 – May 2026
M.S. in Robotics and Autonomous Systems
Focus: Embodied AI, Computer Vision, 3D Scene Understanding, Autonomous Systems
- **Jagran Lakcity University**, Bhopal, India Aug 2019 – Jul 2023
B.Tech. (Hons) CSE (AI)

PROJECTS

ProteinNet-Hybrid — Sequence-to-Structure Inference

- Built a Transformer encoder in PyTorch DDP consuming amino-acid sequence and evolutionary-profile (PSSM) inputs from ProteinNet CASP12, achieving C-alpha coordinate prediction at 10.15 Å validation RMSD after 100 epochs of distributed training.
- Engineered a two-mode inference system (FASTA-only and MSA alignment), packaged into a published Docker image with bundled checkpoint and config for zero-setup CPU inference and reproducible PDB/JSON structure export.
- Evaluated exported predictions through PyMOL overlay inspections across 10 CASP12 targets, yielding an aligned RMSD range of 2.43–15.59 Å and mean of 9.17 Å, confirming reliable coarse fold capture.

Multi-Robot Disaster Management — Safety-Aware Relief Coordination

- Designed a MATLAB multi-robot disaster-response simulator combining weighted-Voronoi coverage control over a spatial priority and time-varying demand field with greedy unique hotspot claiming to eliminate duplicate task assignment across robots.
- Implemented a CBF-style safety controller with QP-based primary filtering and smooth pairwise repulsion fallback, paired with a capacity-aware service model that depletes demand within a service radius and triggers base-return refill cycles.
- Instrumented full metric tracking across locational cost, unmet demand, minimum inter-robot separation, and served coverage over a 180 s scenario, enabling quantitative evaluation of fleet coordination and safety margins.

LERF-TOGO / Gaussian Splatting — Zero-Shot 3D Grasp Semantics

- Extended the LERF-TOGO grasping pipeline with a Gaussian Splatting scene backend abstraction supporting both original LERF/Nerfstudio and splatfacto-based geometry from a unified grasp-generation interface.
- Upgraded scene semantics from CLIP to SigLIP 2 multi-scale pyramid embeddings over HLOC/COLMAP-reconstructed tabletop scenes, with robot-frame alignment to transfer SfM geometry into executable robot coordinates.
- Integrated GraspNet/Dex-Net collision-filtered grasp generation over world-frame point clouds, served through an interactive Viser UI supporting natural-language object-part queries for zero-shot semantic manipulation.

PROFESSIONAL EXPERIENCE

Data Storytelling Assistant — Newswell (Arizona State University), Tempe, Arizona

Nov 2025 – Present

- Built and owned a Python + SQLite ingestion pipeline across 14 public sources from RSS feeds, Reddit, and government agendas, generating daily reporter digests averaging 44 new items per run with peaks of 79 items/day.
- Built OpenAI summarization and outbox delivery modules that assemble local-news and government-meeting digests in 5–6 minute scheduled runs for active editorial review.
- Fixed duplicate-meeting bugs and date-window drift in the Hyland collector with URL normalization, quality-based deduplication, and post-filters, preventing stale or broken updates from resurfacing in reporter emails.
- Added rules-based Reddit relevance scoring and dry-run review checks, reducing noise and making pre-publication editorial QA reproducible before digest release.

Software Engineer — Patel Motors (Indore) Pvt. Ltd., Indore, Madhya Pradesh

May 2023 – May 2024

- Built an ML-assisted Flask + SQL quotation engine that automated client quote generation by learning from historical pricing patterns, replacing manual assembly with faster, data-driven, and fully traceable response handling.
- Rebuilt the company website with HTML, CSS, JavaScript, and Flask-backed forms, integrating dynamic recommendation logic to surface relevant vehicle options based on customer inquiry inputs.
- Designed and deployed a lightweight customer intent classifier on incoming inquiry data, enabling automated lead prioritization and routing to the appropriate sales workflow.

Machine Learning Intern — Value Matrix, Dover, Delaware

Sep 2022 – Nov 2022

- Engineered core components of a real-time eye-movement tracking pipeline using computer vision models, redesigning the evaluation workflow to improve analysis accuracy by 20
- Built an end-to-end review interface integrating AWS Transcribe for automated speech-to-text transcription with a Flask + JavaScript frontend, cutting turnaround time from model output to analyst review significantly.
- Researched and benchmarked multiple gaze estimation architectures to identify the highest-performing model configuration, informing pipeline design decisions and improving overall system reliability.

Associate Intern — Feyn Labs, Gauhati, Assam

Jul 2022 – Sep 2022

- Built EV market-segmentation analyses from customer and competitor data, enabling reports that identified high-potential customer groups for business strategy.

SKILLS

Languages: Python, C++, SQL, JavaScript, MATLAB, C.

Robotics & Frameworks: ROS, PyTorch, OpenCV, Nerfstudio, Gazebo, TensorFlow, TensorFlow Lite, Flask.

AI/ML Concepts: Embodied AI, Gaussian Splatting, Zero-Shot Grasping, Computer Vision, LLMs, Reinforcement Learning, Deep Learning, Autonomous Systems.

Tools & Infrastructure: Git, Docker, AWS, SQLite, Hugging Face, MATLAB Simulink.